

DESIGN OF LOW-POWER HIGH-PERFORMANCE 2–4 AND 4–16 MIXED-LOGIC LINE DECODERS

Abstract:

This brief introduces a mixed-logic design method for line decoders, combining transmission gate logic, pass transistor dual-value logic, and static complementary metal-oxide semiconductor (CMOS). Two novel topologies are presented for the 2–4 decoder: a 14-transistor topology aiming on minimizing transistor count and power dissipation and a 15-transistor topology aiming on high power-delay performance. Both normal and inverting decoders are implemented in each case, yielding a total of four new designs. Furthermore, four new 4–16 decoders are designed by using mixed-logic 2–4 predecoders combined with standard CMOS postdecoder. All proposed decoders have full-swinging capability and reduced transistor count compared to their conventional CMOS counterparts. Finally, a variety of comparative spice simulations at 32nm show that the proposed circuits present a significant improvement in power and delay, outperforming CMOS in almost all cases.

TOOLS:

- Xilinx ISE 14.7

LANGUAGE:

- Verilog HDL

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